# Likelihood and the Probability Scale

Name _____ Date _____ No calculator Version 1

**KEY WORDS** likelihood · impossible · unlikely · even chance · likely · certain · sample space · outcome

## A Likelihood words ●○○ — ANSWER ON THE SHEET

**Remember:** Impossible: it cannot happen. Unlikely: it probably will not happen. Even chance: it is just as likely to happen as not. Likely: it probably will happen. Certain: it must happen.

**1.** Choose the best word for each event: impossible, unlikely, even chance, likely or certain.

**a)** Getting a head when a fair coin is tossed _____

**b)** Rolling a 6 on a fair dice _____

**c)** Rolling a number less than 7 on a fair dice _____

**d)** Rolling a number greater than 1 on a fair dice _____

**e)** Rolling a 9 on a fair dice _____

**f)** Picking a red counter from a bag that holds 1 red and 9 blue counters _____

**g)** Picking a blue counter from the same bag _____

**h)** Rolling an even number on a fair dice _____

## B Ordering by likelihood ●●○ — WORK IN YOUR BOOK

**2.** Put the three events in order, from least likely to most likely. Write the letters.

**a)** A: Rolling a number less than 6 on a fair dice. B: Rolling a 6 on a fair dice. C: Getting a head on a fair coin.

**b)** A: Picking a red counter from a bag of 5 red counters. B: Picking a red counter from a bag of 5 blue counters. C: Picking a red counter from a bag of 2 red and 3 blue counters.

**c)** A: Rolling an even number on a fair dice. B: Rolling a 2 on a fair dice. C: Rolling a number greater than 2 on a fair dice.

## C The probability scale ●●○ — ANSWER ON THE SHEET

**Remember:** A probability is a number from 0 to 1. Impossible is 0, an even chance is $\frac{1}{2}$ and certain is 1.

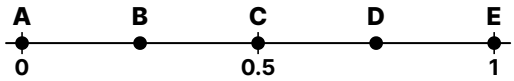
**3.** Write the probability, as a number, of an event that is:

**a)** impossible _____ **b)** certain _____ **c)** an even chance _____



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**4.** Which letter on the scale shows the probability of each event?




**a)** Getting a tail when a fair coin is tossed _____

**b)** Picking a red counter from a bag that holds 1 red and 3 blue counters _____

**c)** Picking a blue counter from the same bag _____

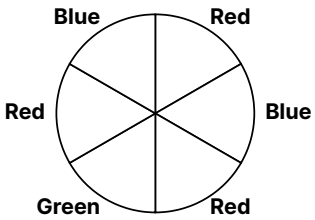
**d)** Rolling a 7 on a fair dice _____

**e)** Rolling a number less than 7 on a fair dice _____

## D Sample space ●●● WORK IN YOUR BOOK

> **Remember:** The sample space is the list of all the possible outcomes. For a coin it is: heads, tails. Be systematic, so that you do not miss any out.

**5.** This spinner has 6 equal parts.




**a)** Write the sample space (every colour it can land on).

**b)** Which colour is it most likely to land on?

**c)** Which colour is it least likely to land on?

**6.** List all the possible outcomes. Use the letters and numbers given.

**a)** A fair dice is rolled.

**b)** A coin is tossed twice (H, T).

**c)** A coin is tossed (H, T) and a spinner numbered 1, 2, 3 is spun.

**7.** How many outcomes are there?

**a)** A coin is tossed

**b)** A dice is rolled

**c)** A coin is tossed and a dice is rolled

**d)** 2 shirts and 3 pairs of shorts

## ★ Extension FOR EARLY FINISHERS

**8.** You choose a sandwich (cheese C, ham H, egg E) and a drink (juice J, water W). List all the possible lunches.

**9.** Give an example of an event that is certain, and an example of an event that is impossible.


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# Likelihood and the Probability Scale

**Name** _____ **Date** _____ No calculator **Version 2**

**KEY WORDS** likelihood · impossible · unlikely · even chance · likely · certain · sample space · outcome

## A Likelihood words ●○○ — ANSWER ON THE SHEET

> **Remember:** Impossible: it cannot happen. Unlikely: it probably will not happen. Even chance: it is just as likely to happen as not. Likely: it probably will happen. Certain: it must happen.

**1.** Choose the best word for each event: impossible, unlikely, even chance, likely or certain.

**a)** Getting a tail when a fair coin is tossed _____

**b)** Rolling a 1 on a fair dice _____

**c)** Rolling a number greater than 0 on a fair dice _____

**d)** Rolling a number less than 6 on a fair dice _____

**e)** Rolling a 0 on a fair dice _____

**f)** Picking a blue counter from a bag that holds 9 red and 1 blue counter _____

**g)** Picking a red counter from the same bag _____

**h)** Rolling an odd number on a fair dice _____

## B Ordering by likelihood ●●○ — WORK IN YOUR BOOK

**2.** Put the three events in order, from least likely to most likely. Write the letters.

**a)** A: Getting a tail on a fair coin. B: Rolling a number greater than 1 on a fair dice. C: Rolling a 3 on a fair dice.

**b)** A: Picking a blue counter from a bag of 4 red counters. B: Picking a blue counter from a bag of 3 blue counters and 1 red counter. C: Picking a blue counter from a bag of 4 blue counters.

**c)** A: Rolling a number less than 3 on a fair dice. B: Rolling an odd number on a fair dice. C: Rolling a 5 on a fair dice.

## C The probability scale ●●○ — ANSWER ON THE SHEET

> **Remember:** A probability is a number from 0 to 1. Impossible is 0, an even chance is $\frac{1}{2}$ and certain is 1.

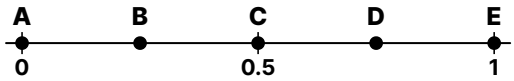
**3.** Write the probability, as a number, of an event that is:

**a)** impossible _____ **b)** certain _____ **c)** an even chance _____



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**4.** Which letter on the scale shows the probability of each event?




**a)** Getting a head when a fair coin is tossed _____

**b)** Picking a green counter from a bag that holds 3 green and 1 red counter _____

**c)** Picking a red counter from the same bag _____

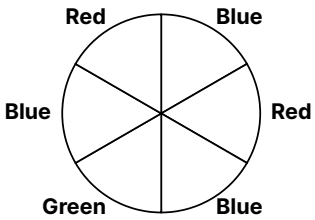
**d)** Rolling an 8 on a fair dice _____

**e)** Rolling a number greater than 0 on a fair dice _____

## D Sample space ●●● WORK IN YOUR BOOK

> **Remember:** The sample space is the list of all the possible outcomes. For a coin it is: heads, tails. Be systematic, so that you do not miss any out.

**5.** This spinner has 6 equal parts.




**a)** Write the sample space (every colour it can land on).

**b)** Which colour is it most likely to land on?

**c)** Which colour is it least likely to land on?

**6.** List all the possible outcomes. Use the letters and numbers given.

**a)** A spinner numbered 1 to 5 is spun.

**b)** A coin is tossed twice (H, T).

**c)** A coin is tossed (H, T) and a spinner numbered 1, 2, 3, 4 is spun.

**7.** How many outcomes are there?

**a)** A coin is tossed

**b)** A dice is rolled

**c)** A coin is tossed and a dice is rolled

**d)** 3 shirts and 2 pairs of shorts

## ★ Extension FOR EARLY FINISHERS

**8.** You choose an ice cream (vanilla V, chocolate C, mint M) and a topping (nuts N, sauce S). List all the possible choices.

**9.** Give an example of an event that is certain, and an example of an event that is impossible.


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